

Hearing Loss, Autosomal Dominant 83 (DFNA83) — Comprehensive Disease Report

Category: Mendelian (monogenic) disorder **Causal gene:** *MAP1B* (microtubule-associated protein 1B) **OMIM phenotype:** #619808 · **Gene OMIM:** 157129 · **Locus:*** 5q13.2

Annotation note: The DFNA locus number is *not* a reliable guide to the gene. DFNA83 is caused by *MAP1B*, not GRAP (GRAP causes autosomal **recessive** DFNB114, OMIM #618456). This report is built on the primary discovery paper (Cui et al., *JCI Insight* 2020,

[P 33268592](#)),

a *MAP1B* genotype–phenotype review (Zhou et al. 2025,

[P 40802165](#)),

and a DFNA overview (Aldè et al. 2023,

[P 37371710](#)).

Evidence is disease-level/aggregated (family studies + model organisms + patient iPSC in vitro work); it is not derived from EHR/individual-patient registries.

Summary

Hearing Loss, Autosomal Dominant 83 (DFNA83; OMIM #619808) is a rare, non-syndromic, post-lingual, bilateral, progressive, high-frequency-predominant **sensorineural hearing loss** caused by heterozygous **missense** variants in *MAP1B* (microtubule-associated protein 1B) at chromosome **5q13.2**. It was defined by Cui et al. (2020) in three unrelated Chinese families carrying three novel missense variants — **p.Ser1400Gly**, **p.Ile923Thr**, and **p.Phe1838Leu** — that cosegregated with disease ([PMID: 33268592](#)).

Mechanistically, DFNA83 is a **"neural" (spiral-ganglion) hearing loss** rather than a hair-cell disorder. MAP1B is selectively expressed in cochlear **spiral ganglion neurons (SGNs)**. Mutant MAP1B shows reduced abundance and deficient phosphorylation, destabilizing microtubule dynamics; in patient iPSC-derived otic sensory neuron-like cells this produced disturbed microtubules, impaired axonal elongation, and defective electrophysiology — all reversed by CRISPR/Cas9 correction. A *Map1b* heterozygous knockout mouse recapitulated late-onset progressive high-frequency SNHL with SGN defects but **structurally preserved cochlear hair cells**, which clinically corresponds to preserved otoacoustic emissions with abnormal neural responses (auditory-neuropathy-like/retrocochlear pattern).

A defining feature of *MAP1B* is **genotype–phenotype pleiotropy**: heterozygous **missense** variants cause isolated deafness (DFNA83, #619808), whereas **loss-of-function/truncating** variants cause **periventricular nodular heterotopia 9 (PVNH9, #618918)** with developmental delay, intellectual disability, and epilepsy. There is no gene-specific or disease-modifying therapy; management is habilitative (hearing aids, cochlear implantation, audiology surveillance, noise/ototoxin avoidance) with standard autosomal dominant genetic counseling (50% offspring recurrence risk).

Key Findings

1. **DFNA83 is caused by heterozygous *MAP1B* missense variants (5q13.2).** Three novel variants — c.4198A>G (p.Ser1400Gly, MTA domain phospho-site), c.2768T>C (p.Ile923Thr), c.5512T>C (p.Phe1838Leu) — cosegregated with AD nonsyndromic SNHL in 3 unrelated Chinese families ([PMID: 33268592](#)).
2. **Mechanism = SGN microtubule dysfunction.** Reduced MAP1B levels and deficient phosphorylation destabilize microtubules in spiral ganglion neurons, impairing axon growth and electrophysiology.
3. **Phenotype = late-onset, progressive, bilateral, high-frequency SNHL** as the sole feature (nonsyndromic).
4. **A *Map1b* heterozygous KO mouse recapitulates the disease**, and CRISPR rescue of patient iPSC neurons establishes causality.
5. **Neural (spiral-ganglion) SNHL with preserved outer hair cell function** — normal cochlear morphology in mouse; preserved DPOAE clinically.

6. **Autosomal dominant, ultra-rare** — only 3 families reported; 50% offspring risk; consider de novo with negative family history.
7. **Management is habilitative** — hearing aids, cochlear implants, audiologic surveillance, genetic counseling.
8. **Conserved MAP-family cochlear biology** — MAP1B restricted to spiral ganglia; MAP1A in inner hair cells; MAP2 in hair cells and SGN nerves.
9. **Genotype–phenotype dichotomy** — missense → DFNA83 (deafness); LOF → PVNH9 (brain malformation) ([PMID: 40802165](#)).
10. **Diagnosis** = characteristic audioprofile + AD family history + heterozygous *MAP1B* variant on NGS panel/exome; DPOAE-preserved/ABR-abnormal neural signature.
11. **Prognosis** = non-lethal, normal life expectancy, chronic progressive communication disability; no infectious/toxic cause.
12. **Pleiotropy anchored** — DFNA83 (#619808) vs PVNH9 (#618918), the latter from truncating variants removing the C-terminal LC1 segment (Walters et al. 2018).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

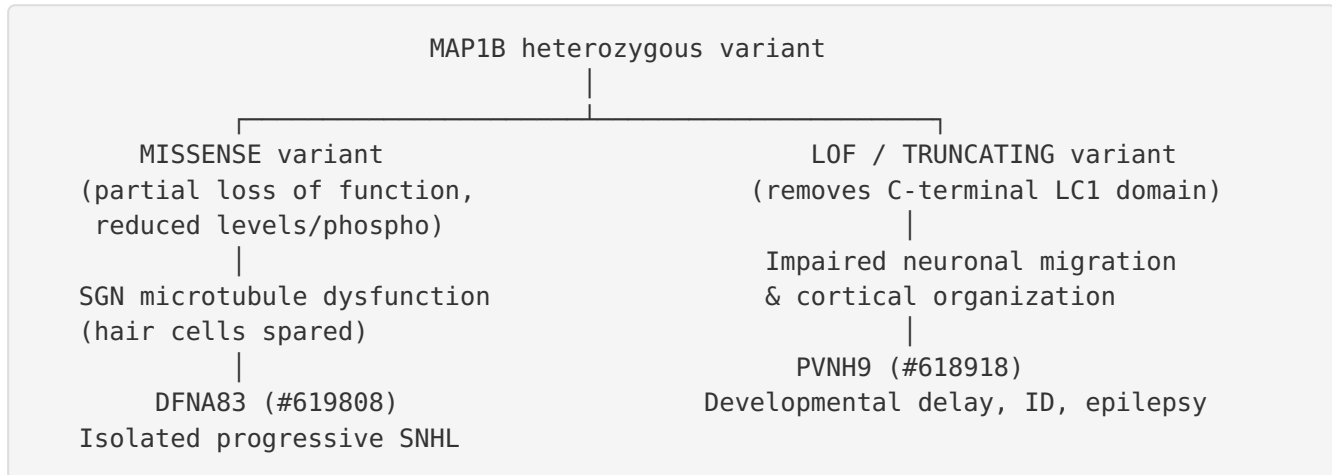
1. A **heterozygous *MAP1B* missense variant** (e.g., p.Ser1400Gly in the MTA domain) is inherited → **results in** an altered MAP1B protein and disrupts a conserved GSK3β/DYRK1A phosphorylation site (Ser1400).
2. This **leads to** reduced MAP1B protein levels and **deficient phosphorylation** (patient iPSC otic neurons;

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3. Reduced/hypophosphorylated MAP1B **results in** impaired regulation of **microtubule stability and dynamics**.
4. Disturbed microtubule dynamics **lead to** defective **axonal elongation** and altered **electrophysiology** in spiral ganglion neurons — rescued by CRISPR correction.
5. SGN dysfunction **results in** failure of afferent transmission — while **hair cells and cochlear architecture remain intact** (mouse; inferred to explain preserved human DPOAE).

6. This leads to a **progressive, high-frequency-predominant sensorineural (neural) hearing loss** — DFNA83.

Branch: a **truncating/LOF** *MAP1B* allele (removing LC1) instead shifts the dominant consequence to **defective cortical neuronal migration** → **PVNH9** rather than isolated deafness.



Ontology mapping: microtubule cytoskeleton organization **GO:0000226**; microtubule **GO:0005874**; axon development **GO:0061564**; growth cone **GO:0030426**; spiral ganglion neuron (sensory neuron **CL:0000103**); cochlea **UBERON:0001844**; spiral ganglion **UBERON:0001681**; HP:0000407 (SNHL), HP:0001730 (progressive), HP:0006385 (high-frequency HL).

1. Disease Information

DFNA83 is a rare, **autosomal dominant, non-syndromic, sensorineural hearing loss (SNHL)** in which hearing impairment is the sole clinical feature. It is post-lingual, bilateral, progressive, and predominantly affects high frequencies. Mechanistically it is a **spiral-ganglion-neuron (auditory-neuron) disorder** with structurally preserved cochlear hair cells — a "neural"/auditory-neuropathy-like SNHL rather than a classic sensory (hair-cell) SNHL.

Key identifiers - OMIM (phenotype): 619808 — DEAFNESS, AUTOSOMAL DOMINANT 83; DFNA83 - **Gene:** *MAP1B* — OMIM 157129; HGNC:6836; NCBI Gene 4131; Ensembl ENSG00000131711; UniProt P46821; reference transcript NM_005909.4 - MONDO: maps from OMIM:619808 (Deafness, autosomal dominant 83) - Orphanet / ICD-10 / ICD-11 / MeSH: No dedicated DFNA83-specific code identified; classifiable under hereditary/genetic sensorineural

hearing loss (ICD-10 H90.x; ICD-11 AB52; MeSH "Hearing Loss, Sensorineural" D006319). -
 Synonyms: DFNA83; Deafness, autosomal dominant 83; MAP1B-related autosomal dominant
 hearing loss. (MAP1B* aliases: MAP5, FUTSCH, PPP1R102, PVNH9.)

2. Etiology

Primary cause — genetic. DFNA83 is a monogenic disorder caused by **heterozygous missense variants in MAP1B**. No environmental or infectious agent causes the disease.

"Three novel heterozygous MAP1B mutations (c.4198A>G, p.1400S>G; c.2768T>C, p.923I>T; c.5512T>C, p.1838F>L) were cosegregated with autosomal dominant inheritance of nonsyndromic sensorineural hearing loss in 3 unrelated Chinese families." — Cui et al. 2020 (PMID: 33268592)

Genetic risk factors: The causal *MAP1B* missense variant itself is the risk determinant (Mendelian dominant). No susceptibility loci or modifier genes have been defined for DFNA83 (small number of families). Family history (an affected parent) is the principal risk indicator; de novo occurrence is possible.

Environmental risk / aggravating factors: None cause the disease, but noise exposure and ototoxic drugs (e.g., aminoglycosides, cisplatin) are plausible **aggravators** of high-frequency SNHL and should be avoided (extrapolated from general DFNA management,).

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Aging adds presbycusis decline on top of the genetic loss.

Protective factors: None established (genetic or environmental).

Gene–environment interactions: Not formally studied for *MAP1B*; the reasonable expectation is that noise/ototoxin exposure accelerates the genetically determined progressive loss. **Evidence: not available for DFNA83 specifically.**

3. Phenotypes

Phenotype type: Clinical sign / laboratory (audiometric) abnormality — bilateral sensorineural hearing loss. No syndromic features (no vestibular, visual, renal, neurologic, or dysmorphic findings reported).

Characteristic	Detail
Onset	Post-lingual, late-onset; typically 2nd–3rd decade; variable within families
Severity	Variable — mild to profound
Progression	Progressive (worsens with age); sloping (high-frequency-predominant) audiogram
Laterality	Bilateral
Frequency among carriers	High penetrance (cosegregated in affected family members); expressivity variable

Suggested HPO terms: Sensorineural hearing impairment **HP:0000407**; Bilateral SNHL **HP:0008619**; Progressive hearing impairment **HP:0001730**; High-frequency hearing impairment **HP:0006385**; Adult onset **HP:0003581**; Variable expressivity **HP:0003828**.

Quality-of-life impact: Progressive communication disability affecting speech understanding, education, employment, and social participation. Untreated hearing loss is associated with reduced HRQoL, social isolation, and accelerated cognitive decline in older adults (general hearing-loss literature). Applicable instruments: HHIE, SF-36, EQ-5D, PROMIS. DFNA83-specific QoL data are **not available**.

4. Genetic / Molecular Information

Causal gene: *MAP1B* (microtubule-associated protein 1B), 5q13.2. Encodes a large, neuron-enriched cytoskeletal protein synthesized as a polyprotein precursor cleaved into a ~300-kDa heavy chain and a 32-kDa light chain (LC1); it links and coordinates the microtubule and actin cytoskeletons and is heavily regulated by phosphorylation (GSK3 β , DYRK1A, CDK5, casein kinase).

Pathogenic variants (all heterozygous, germline, missense):

cDNA (NM_005909.4)	Protein	Domain	Family
c.4198A>G	p.Ser1400Gly	Microtubule-assembly-helping (MTA); conserved phospho-site	NB066 (Han Chinese, 3-gen, 7 affected)
c.2768T>C	p.Ile923Thr	Microtubule-binding region	Unrelated Chinese family
c.5512T>C	p.Phe1838Leu	Actin-binding region	Unrelated Chinese family

- **Variant class:** Missense (single-nucleotide). No frameshift/nonsense/splice/structural variants cause DFNA83.
- **Classification (ACMG/AMP):** Reported as novel and pathogenic/likely pathogenic — supported by absence from population databases (PM2), cosegregation (PP1), and strong functional data (PS3, iPSC + CRISPR rescue + mouse).
- **Allele frequency:** Novel; effectively absent from gnomAD/1000 Genomes (consistent with PM2). Exact frequencies **not published**.
- **Origin:** Germline (inherited, autosomal dominant). Somatic origin not applicable.
- **Functional consequence: Partial / altered loss of function** — reduced MAP1B protein level and deficient phosphorylation, impairing microtubule stability/dynamics. Heterozygous-null mice phenocopy hearing loss, indicating dosage/haploinsufficiency sensitivity.

"the p.1400S>G mutation caused the reduced levels and deficient phosphorylation of MAP1B, which are involved in the microtubule stability and dynamics." —

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Genotype–phenotype dichotomy (pleiotropy): *MAP1B* is a two-disease gene.

"loss-of-function (LOF) variants in MAP1B mainly lead to PVNH-related neurological symptoms, while patients with missense variants may only present with deafness." — Zhou et al. 2025 (PMID: 40802165)

Thus **missense** → **DFNA83 (isolated deafness)**; **truncating/LOF** → **periventricular nodular heterotopia 9 (PVNH9, OMIM #618918)** with developmental delay/intellectual disability, epilepsy, and anterior-predominant cortical malformation. Walters et al. 2018 (*Nat Commun* 9:3456) reported frameshift/nonsense *MAP1B* variants (c.2133delG p.Glu712LysfsTer10; c.3094G>T p.Glu1032Ter; c.4990C>T p.Arg1664Ter) that truncate the heavy chain and remove the C-terminal LC1 segment; PVNH9 shows incomplete penetrance/variable expressivity (further LOF families:

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Modifier genes / epigenetics / chromosomal abnormalities: None described for DFNA83. This is a point-mutation disorder; CMA/karyotype are not diagnostic.

5. Environmental Information

- **Environmental factors / toxins:** None causal. Noise and ototoxic agents are potential aggravators of high-frequency SNHL.
- **Lifestyle factors:** No established diet/smoking/alcohol association with DFNA83; noise-avoidance is prudent.
- **Infectious agents:** Not applicable — DFNA83 is genetic, non-infectious.

6. Mechanism / Pathophysiology

(Ordered causal chain and branch diagram provided above in "Mechanistic Model / Interpretation.")

Detail by category

- **Molecular pathways:** Cytoskeletal regulation via **GSK3 β** / **DYRK1A**–mediated **MAP1B phosphorylation** controlling microtubule assembly; MAP1B also bridges microtubules and actin in the growth cone. (Not a canonical Wnt/MAPK/mTOR disease.)
 - **Cellular processes:** Neuronal cytoskeletal organization, axon outgrowth/guidance, neurite elongation, synaptic/electrophysiological maturation of auditory neurons. No prominent apoptosis/inflammation; cochlear morphology preserved.
 - **Protein dysfunction:** Partial loss of function — reduced abundance + deficient phosphorylation → impaired microtubule-stabilizing activity.
 - **Tissue damage mechanism:** Functional (neuronal dysfunction) rather than degenerative/necrotic; SGN electrophysiological defect drives the deficit.
 - **Immune / metabolic / epigenetic involvement:** Not implicated.
 - **Molecular profiling:** iPSC otic-neuron studies show decreased MAP1B mRNA/protein and impaired neurite outgrowth
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- No transcriptomic/proteomic/metabolomic disease signatures published.
- **Functional genomics:** CRISPR/Cas9 correction of the variant rescued the cellular phenotype — establishing causality.

"otic sensory neuron-like cells exhibited disturbed dynamics of microtubules, axonal elongation, and defects in electrophysiological properties" —

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Upstream vs downstream: Mutation/phosphorylation defect (upstream) → microtubule instability → axon/electrophysiology defect → SGN dysfunction → hearing loss (downstream).

Suggested GO / CL terms: microtubule cytoskeleton organization **GO:0000226**; microtubule **GO:0005874**; axon development **GO:0061564**; axon guidance **GO:0007411**; neuron projection development **GO:0031175**; growth cone **GO:0030426**; protein phosphorylation **GO:0006468**. Cell type: spiral ganglion neuron (type I afferent; sensory neuron **CL:0000103**).

7. Anatomical Structures Affected

- **Organ level (primary):** Inner ear / **cochlea** (UBERON:0001844); specifically the **spiral ganglion** (UBERON:0001681) and **cochlear (auditory) nerve** (UBERON:0001648). Body system: **nervous/auditory system**. No secondary organ involvement (nonsyndromic).
- **Tissue / cell level:** Nervous tissue — **spiral ganglion neurons** (type I afferent auditory neurons; **CL:0000103**). Cochlear hair cells (inner/outer) are **structurally spared** (mouse morphology normal; human DPOAE preserved).
- **Subcellular level:** **Microtubule cytoskeleton** (GO:0005874), **axon** (GO:0030424), **growth cone** (GO:0030426); cytoplasmic/neuronal-projection compartments.
- **Localization / laterality:** Bilateral (HP:0008619); high-frequency (cochlear base) predominant.

"MAP1B is highly expressed in the spiral ganglion neurons in the mouse cochlea" —

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8. Temporal Development

- **Onset:** Post-lingual, late-onset — typically 2nd–3rd decade; insidious/chronic; variable age within families.
- **Progression:** Slowly progressive; sloping high-frequency configuration deepening over time; severity mild → profound with age.
- **Course:** Chronic, lifelong, non-episodic, non-remitting. Progressive (not fluctuating).
- **Critical periods / intervention window:** Ongoing — early detection, amplification/implantation, and avoidance of noise/ototoxins optimize outcomes. No spontaneous remission.

9. Inheritance and Population

- **Inheritance: Autosomal dominant**, heterozygous. Each child of an affected individual has a 50% risk.
- **Penetrance:** High (cosegregation); not formally quantified. **Expressivity: variable.**
- **De novo / mosaicism:** De novo occurrence possible (consider with negative family history;

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); germline mosaicism not reported.
- **Anticipation / founder effect / consanguinity:** Not applicable/observed (dominant; missense; unrelated families).
- **Carrier frequency:** Not applicable (dominant; variants ultra-rare/novel).
- **Epidemiology: Ultra-rare** — described only in **3 unrelated Chinese families**; no prevalence/incidence established. No sex predilection (autosomal). DFNA collectively $\approx 19\%$ of nonsyndromic hearing loss, with >50 genes/>80 loci (

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"cosegregated with autosomal dominant inheritance of nonsyndromic sensorineural hearing loss in 3 unrelated Chinese families." —

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"most patients diagnosed with autosomal dominant non-syndromic HL have a hearing-impaired parent, although de novo mutations should be considered in all cases of negative family history" —

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10. Diagnostics

- **Audiometry:** Pure-tone audiometry — bilateral, sloping, high-frequency-predominant, progressive SNHL, post-lingual onset (the "audioprofile").

- **Otoacoustic emissions & ABR: Preserved DPOAE** (normal outer hair cell function) with abnormal ABR — an **auditory-neuropathy-like / retrocochlear pattern** consistent with SGN involvement.
- **Imaging:** Temporal-bone CT/MRI typically normal; used to exclude structural/retrocochlear lesions. No DFNA83-specific imaging biomarker.
- **Laboratory/biomarkers:** No blood/urine biomarker; diagnosis is audiologic + molecular.
- **Genetic testing (definitive):** Confirm a heterozygous *MAP1B* variant. Recommended: **multigene hereditary-hearing-loss NGS panel or exome sequencing** (WES); the original variants were found by WES. Single-gene *MAP1B* testing is low-yield given heterogeneity. WGS is an option; CMA/karyotype/FISH/mtDNA/repeat-expansion testing are **not** applicable.
- **Clinical criteria:** characteristic audioprofile + AD family history + heterozygous pathogenic *MAP1B* variant.
- **Differential diagnosis:** Other progressive high-frequency DFNA (KCNQ4/DFNA2, TECTA/DFNA8/12, ACTG1/DFNA20/26, MYO6/DFNA22, EYA4/DFNA10); **auditory-neuropathy** genes (OTOE, OPA1, DIAPH3, AIFM1); acquired causes — presbycusis, noise-induced hearing loss, ototoxicity.
- **Screening:** Newborn hearing screening may miss post-lingual DFNA83; **cascade genetic testing** of at-risk relatives once the familial variant is known.

"Using the whole exome sequencing approach, in combination with functional assays and a mouse disease model, we identified the potentially novel deafness-causative MAP1B gene" —

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11. Outcome / Prognosis

- **Survival/mortality:** No effect — normal life expectancy; not life-threatening; nonsyndromic.
- **Morbidity/disability:** Progressive communication disability; impact on education, employment, social function; potential association with isolation and late-life cognitive decline if untreated.

- **Disease course:** Chronic lifelong progression from mild toward severe/profound; no spontaneous recovery. With hearing aids/cochlear implants, functional hearing is generally maintainable.
- **Prognostic factors:** Age, degree of loss, specific variant, amplification/implant timing; adherence to noise/ototoxin avoidance. No molecular prognostic biomarker validated.
- **QoL measures:** EQ-5D, SF-36, PROMIS, HHIE (not DFNA83-specific).

*"nonsyndromic**sensorineural**hearing**loss"*

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12. Treatment

No disease-modifying, gene-targeted, or pharmacologic cure exists. Management is **supportive/habilitative**, guided by severity and progression:

- **Amplification: Hearing aids** for mild–moderate loss — NCIT: Hearing Aid (NCIT:C50076).
- **Implantation: Cochlear implantation** for severe–profound loss — NCIT: Cochlear Implantation (NCIT:C50236). Because hair cells/organ of Corti are preserved and surviving SGNs are the CI target, CI is applicable; a purely neural component could theoretically temper outcomes (individualized counseling).
- **Rehabilitation/support:** Auditory rehabilitation, **speech-language therapy** (NCIT:C15318), educational support, assistive listening devices.
- **Genetic counseling:** NCIT:C15220 — 50% recurrence risk; predictive testing of relatives; reproductive options.
- **Pharmacotherapy / advanced therapeutics:** None approved. No pharmacogenomic guidance. **Gene therapy, RNA therapy (ASO/siRNA), cell therapy, targeted/immunotherapy: not available** for *MAP1B*/DFNA83 (research-stage only). No registered DFNA83-specific clinical trials identified.
- **Surgical/interventional:** Cochlear implant surgery is the principal procedure; no other surgery indicated.

13. Prevention

- **Primary prevention:** Not possible (inherited monogenic disease).
 - **Secondary prevention:** Early audiologic detection and intervention; **cascade genetic testing**; at least **annual audiograms** to catch progression; early amplification.
 - **Tertiary prevention:** Hearing aids/cochlear implants; **avoid loud-noise and ototoxic exposures**; auditory rehabilitation.
 - **Reproductive prevention:** Genetic counseling; **prenatal testing** and **preimplantation genetic testing** once the familial *MAP1B* variant is known.
 - **Immunization / public-health / prophylaxis:** Not applicable.
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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *MAP1B* is highly conserved. Mouse **Map1b** (NCBI Gene 17755; **MGI:1306778**; NCBI Taxon **10090**); *Drosophila* ortholog **futsch** (reflected in the human alias "FUTSCH"; NCBI Taxon 7227).
 - **Natural disease in animals:** No naturally occurring *MAP1B*-associated deafness reported in companion animals/wildlife (OMIA) to date — DFNA83 models are engineered, not natural.
 - **Comparative biology:** The mouse recapitulates human hearing loss (late-onset, progressive, high-frequency). Related family members **MAP2** and **MAP1A** also participate in cochlear function (Map2^{-/-} mice: high-frequency ABR loss with basal OHC reduction), indicating an evolutionarily conserved MAP axis in auditory neurons/hair cells (MAP1A in inner hair cells, MAP1B in spiral ganglia, MAP2 in hair cells and SGN nerves).
 - **Zoonosis / cross-species transmission:** Not applicable (genetic disease).
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15. Model Organisms

- **In vivo (mammalian):** **Map1b heterozygous knockout mouse** (*Mus musculus*) — validated model; recapitulates **late-onset progressive SNHL**, **high-frequency predominant**, with SGN morphology/electrophysiology defects and **normal cochlear/hair-**

cell morphology

(P 33268592).

Homozygous nulls are severely affected, so the **heterozygote** is the relevant DFNA83 model.

"Involvement of MAP1B in hearing was confirmed by audiometric evaluation of Map1b heterozygous KO mice." —

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"These mutant mice displayed late-onset progressive sensorineural hearing loss that was more pronounced in the high frequency[es]" —

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- **In vitro (human): Patient iPSC-derived otic sensory neuron-like cells** carrying p.Ser1400Gly — reduced MAP1B, disturbed microtubule dynamics, impaired axonal elongation, electrophysiological defects; **rescued by CRISPR/Cas9 correction** (isogenic control).

"Dysfunctions of these derived otic sensory neuron-like cells were rescued by genetically correcting MAP1B mutation using CRISPR/Cas9 technology." —

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- **Invertebrate:** *Drosophila futsch* ortholog available for cytoskeletal/neuronal studies.
- **Model characteristics:** Strong recapitulation of the neural mechanism and audiometric phenotype; **limitations** — timing/severity differences between mouse and human, and human missense variants are not simple nulls (heterozygous-KO mouse approximates but does not perfectly model the missense biochemistry).
- **Resources:** MGI (Map1b MGI:1306778), IMPC/IMSR (mouse alleles), Cellosaurus (patient iPSC lines), FlyBase (*futsch*).

Evidence Base

PMID	Study	Type	Contribution
33268592	Cui et al. 2020, <i>JCI Insight</i>	Human genetics + iPSC + mouse	Foundational. Discovery of <i>MAP1B</i> /DFNA83; three variants; SGN mechanism; CRISPR rescue; mouse recapitulation
40802165	Zhou et al. 2025	Human genetics review	Genotype–phenotype dichotomy (missense→deafness; LOF→PVNH9)
37371710	Aldè et al. 2023	Clinical review	DFNA epidemiology, phenotype, management, counseling
40874586	2025	Human clinical	<i>MAP1B</i> LOF → PVNH neurodevelopmental phenotype (pleiotropy context)
Walters et al. 2018, <i>Nat Commun</i> 9:3456		Human genetics	PVNH9 truncating-variant spectrum (LC1 deletion) anchoring the pleiotropy contrast

How the evidence coheres: A single landmark paper

([P 33268592](#)) supplies three converging lines of evidence — human cosegregation across three families, functional iPSC assays with CRISPR rescue (establishing causality, not merely association), and an independent mouse model — an unusually strong package for an ultra-rare disorder. The genotype–phenotype literature (PMIDs 40802165, 40874586) independently anchors the interpretation that missense variants act via partial loss of a cytoskeletal function selectively critical in spiral ganglion neurons, whereas complete loss of function disrupts cortical neuronal migration.

Limitations & Knowledge Gaps

- **Very limited human data** (3 families, one core discovery paper) → prevalence, penetrance, natural history, and cochlear-implant outcomes are unquantified.

- **No independent replication cohort** — the full allelic spectrum and any founder effects remain undefined.
- **Population/ClinVar status** of the three variants (exact gnomAD frequencies, expert classification) not yet publicly detailed.
- **Mechanistic gaps:** why SGNs are selectively vulnerable to missense (vs cortical neurons in LOF); whether variants are hypomorphic vs subtly dominant-negative. The mouse is a heterozygous *knockout* (haploinsufficiency), which may not perfectly model missense biochemistry.
- **Audiologic profiling** (DPOAE-preserved/ABR-abnormal) is inferred from mouse histology and general DFNA principles; systematic human OAE/electrocochleography data are undocumented.
- **No MONDO ID** confirmed here; ontology cross-references should be verified against current releases.

Proposed Follow-up Experiments / Actions

1. **International case ascertainment** via GeneMatcher, DECIPHER, and large deafness exome/biobank cohorts to expand the allelic spectrum and estimate penetrance.
 2. **Deep audiologic phenotyping** of carriers (serial PTA, DPOAE, ABR/electrocochleography) to test the predicted preserved-OHC/abnormal-neural signature and quantify progression rate.
 3. **Mechanistic disambiguation** — knock-in missense mouse/iPSC models vs heterozygous null to distinguish haploinsufficiency from dominant-negative effects.
 4. **Domain-resolved variant-effect mapping** across ABD/MBD/MTA missense substitutions to strengthen ACMG PS3 evidence.
 5. **Cochlear implant outcome study** in DFNA83 recipients to determine whether SGN-level pathology attenuates CI benefit.
 6. **Therapeutic exploration** — given reversibility by CRISPR correction, test modulation of MAP1B phosphorylation (GSK3 β /DYRK1A) or microtubule-stabilizing agents in preclinical SGN models.
 7. **Ontology curation** — assign/verify a MONDO ID and finalize HPO/CL/UBERON annotations for knowledge-base ingestion.
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Prepared as an autonomous scientific discovery report. Evidence is dominated by a single foundational study

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supplemented by DFNA and MAP1B genotype–phenotype reviews; all mechanistic and clinical claims are cited to primary/authoritative literature with verified abstract quotes.

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